

Global Issues in Engineering Ethics

Introduction to Global Ethics

- Globalization integrates nations via trade, technology, and culture.
- Engineers and MNCs play key roles in globalization.
- Ethical considerations include business, environment, and technology.

Multinational Companies (MNCs)

- MNCs operate in home and host countries.
- Influence local and global economies.
- Responsibilities: Wages, safety, employee benefits, ethical operations.

Business Ethics

- Set of moral guidelines for business conduct.
- Includes employee welfare, environmental care, waste management.
- Promotes trust, transparency, and social good.

Environmental Ethics

- Focus on moral issues affecting the environment.
- Pollution, resource depletion, global warming are major concerns.
- Engineers must develop eco-friendly solutions and sustainable designs.

Computer Ethics

- Concerns with privacy, property, consent, and social harm.
- Ten Commandments of Computer Ethics include avoiding harm, respecting property, and ensuring honesty.
- Growth of cybercrime is an ethical challenge.

Computers in Technological Development

- Computers enhance communication and commerce.
- Issues: Free speech, financial fraud, software theft, identity theft.
- Geographical coverage makes tracking cybercrimes difficult.

Privacy Factors

- Cyber threats risk personal and corporate data.
- Viruses, hacking, data breaches compromise security.
- Legal protections like the Privacy Act of 1947 exist.

Weapons Development

- Engineering contributes to modern weapons.
- Ethical dilemma: National defense vs. human harm.
- Engineers may feel conflicted about military applications.

Engineers as Managers

- Engineers must resolve conflicts and promote ethical policies.
- Balance between profitability and social responsibility.
- Ensure safe and valuable product development.

Managing Conflicts

- Conflicts arise from scheduling, priorities, resources, and personalities.
- Managers must resolve conflicts with patience and ethical focus.
- Objective: Efficiency, quality, customer satisfaction.

Consulting Engineers

- Provide specialized knowledge independently.
- Maintain ethical clarity in consulting and avoid conflicts of interest.
- Offer unbiased expert advice.

Engineers as Advisors

- Must recognize limitations and work with qualified experts.
- Adopt roles: Hired Guns, Value-Neutral Analysts, Value-Guided Advocates.
- Honesty and public welfare must guide decisions.

Conclusion

- Global ethics demand responsibility in engineering.
- Balancing innovation, profit, and social good is key.
- Continuous ethical practice ensures long-term success.